- $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields
- But $\mathbb{Q}$ has its problems:

<u>Prop</u> $\sqrt{2}$ is irrational. I.e., $\nexists\, q \in \mathbb{Q}$ s.t. $q^2 = 2$.

<u>Pf</u>: Suppose $\exists\, q \in \mathbb{Q}$ s.t. $q^2 = 2$. Then we may write $q = \frac{a}{b}$ for $a \in \mathbb{Z}$ and $b \in \mathbb{N}$ with no common factors.

Then

$$2 = q^2 = \frac{a^2}{b^2} \implies a^2 = 2b^2 \implies a^2 \text{ even}$$

$$\implies a \text{ even} \implies a^2 = 2b^2 \text{ is divisible by } 4$$

$$\implies b^2 \text{ is even} \implies b \text{ is even.}$$

So a and b are both divisible by 2. But this contradicts that a and b have no common factors. So no such $q = \frac{a}{b}$ exists. $\square$

- We can recast this problem in terms of sets:

<u>Ex</u> $A = \{x \in \mathbb{Q} : x^2 < 2\}$ is bounded.

2 is an upper bound for A.

A does not have a maximum:

Claim: If $x \in A$, then $\exists\, y \in A$ s.t. $y > x$.

Fix $x \in A$. Set $y = x + \frac{2 - x^2}{2 + x}$. Then:

- $y \in \mathbb{Q}$: since

$$\left.\begin{array}{l} x \in \mathbb{Q} \implies 2 - x^2,\ 2 + x \in \mathbb{Q} \\ x \in A \implies 2 + x > 0 \end{array}\right\} \implies y = x + \underbrace{\frac{2 - x^2}{2 + x}}_{\in \mathbb{Q}} \in \mathbb{Q}$$

- $y > x$: since

$$x \in A \Rightarrow \left\{ \begin{array}{l} 2 - x^2 > 0 \\ 2 + x > 0 \end{array} \right\} \Rightarrow y = x + \underbrace{\frac{2-x^2}{2+x}}_{>0} > x$$

- $y^2 < 2$: since

$$y^2 = \left(\frac{2x + \cancel{x^2} + 2 - \cancel{x^2}}{2+x} \right)^2 = \frac{4(x+1)^2}{(2+x)^2} = \frac{4x^2+8x+4}{x^2+4x+4}$$

$$= \frac{2(x^2+4x+4) + 2x^2 - 4}{x^2+4x+4}$$

$$= 2 + \underbrace{\frac{2(x^2-2)}{(x+2)^2}}_{<0}$$

Therefore $y \in A$ and $y > x$.

Claim: A has no least upper bound in $\mathbb{Q}$.

Suppose $L \in \mathbb{Q}$ is a least upper bound for A. Then $L^2 \not< 2$ by the previous claim ($L^2 < 2 \Rightarrow L \in A \Rightarrow \exists y \in A$ s.t. $y > L \Rightarrow L$ is not an upper bound). Also, $L^2 \neq 2$ by the previous proposition. So $L^2 > 2$. Set $y = L + \frac{2-L^2}{2+L}$. Then, similar to the previous claim, we have:

- $y < L$: since $y = L + \underbrace{\frac{2-L^2}{2+L}}_{<0}$

- $y^2 > 2$: since $y^2 = 2 + \underbrace{\frac{2(L^2-2)}{(L+2)^2}}_{>0} > 2$

- y is an upper bound for A:

$$a > y \Rightarrow a^2 > y^2 > 2 \Rightarrow a \notin A$$

So $a \leq y \ \forall a \in A$.

Therefore L is not the least upper bound.

<u>Def</u> Let $<$ be an order relation on a set F. We say that F has the <u>least upper bound property</u> if it satisfies: $\forall$ nonempty $A \subseteq F$ that is bounded above, $\exists$ a least upper bound $\sup A \in F$.

<u>Ex</u> $(\mathbb{Q}, <)$ does not satisfy the least upper bound property. By the previous example, $A = \{x \in \mathbb{Q} : x^2 < 2\}$:

- is nonempty: $0 \in A$
- is bounded above: by 2
- does not have a least upper bound $L \in \mathbb{Q}$.

<u>Thm</u> $\exists$ an ordered field that has the least upper bound property. We denote it by $(\mathbb{R}, +, \cdot, <)$. This field is unique. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

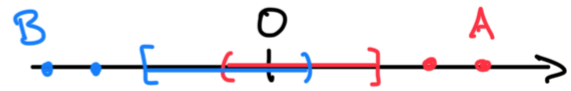
- See the Appendix of Ch. 1 in Rudin for details

<u>Prop</u> $\forall A \subseteq \mathbb{R}$ that is nonempty and bounded below, $\exists$ a greatest lower bound $\inf A \in \mathbb{R}$.

<u>Pf</u>: Fix $A \subseteq \mathbb{R}$ nonempty and bounded below.

Set

$$B = \{-a : a \in A\}.$$



Then:

- $B \neq \emptyset$: As $A \neq \emptyset$, $\exists\, a_0 \in A$. So $-a_0 \in B$.
- B is bounded above: As A is bounded below, $\exists\, m \in \mathbb{R}$ s.t.

$$a \geq m \quad \forall a \in A \quad \Rightarrow \quad -a \leq -m \quad \forall a \in A$$
$$\Rightarrow \quad b \leq -m \quad \forall b \in B$$

So $-m$ is an upper bound for B.

So, by the least upper bound property, $\exists \sup B \in \mathbb{R}$.

Claim: $-\sup B$ is the greatest lower bound for A.

- $-\sup B$ is a lower bound for A:

  $\sup B$ is an upper bound for B

  $\Rightarrow -a \leq \sup B \quad \forall a \in A \quad \Rightarrow \quad a \geq -\sup B \quad \forall a \in A.$
- If m is a lower bound for A then $-\sup B \geq m$:

  m lower bound for $A \Rightarrow a \geq m \quad \forall a \in A$

  $\Rightarrow -a \leq -m \quad \forall a \in A \quad \Rightarrow \quad -m$ upper bound for B

  $\Rightarrow \sup B \leq -m \quad \Rightarrow \quad -\sup B \geq m.$

So $\inf A$ exists and is equal to $-\sup B$. $\square$

<u>Rmk</u> We just proved: If $A \subseteq \mathbb{R}$ is nonempty and bounded below, then $\inf A = -\sup(-A)$.